

Homework for Project Reactor

1. Create Your First Flux and Mono

- Create a `Flux<String>` with 5 fruit names and a `Mono<String>` with your favorite fruit.
- Subscribe to both and print the items.

2. Full Subscribe with Callbacks

Use the fruits Flux and subscribe using the full version:

`onNext`, `onError`, and `onComplete` handlers.

3. Lifecycle Hooks

Using the same fruits Flux, add these hooks in order:

- `doOnSubscribe`
- `doOnNext`
- `doOnComplete` (or `doFinally`)

Print a clear message for each hook so the execution order is visible.

4. Map & Filter

Given `Flux.just(1, 2, 3, 4, 5, 6, 7, 8, 9, 10)`

- Use `map()` to get square of each number
- Use `filter()` to keep only even squares
- Print the result

5. flatMap()

- Create a Flux of names: `["Alice", "Bob", "Charlie"]`
- For each name, use `flatMap()` to return a new Flux that emits the name and its length (2 items per name).
- Example output: `Alice, 5, Bob, 3, ...`

6. concat() vs merge()

Create two Fluxes:

- `Flux1`: 1, 2, 3 (with 500ms delay between items)
- `Flux2`: 10, 20, 30 (with 300ms delay)

a) Combine them using `concat()`

b) Combine them using `merge()`

7. zip()

Create:

- `names = Flux.just("Alice", "Bob", "Charlie")`
- `scores = Flux.just(85, 92, 78)`

Use `zip()` to create a combined output like: `"Alice scored 85"`

8. Error Handling Operators

Create a Flux: `Flux.just("apple", "banana", "cherry")`

Inside `map()`, throw a `RuntimeException` when the item is `"banana"`.

Solve using:

- `onErrorReturn("Default Fruit")`
- `onErrorResume()` → return a fallback Flux with 2 fruits
- `onErrorMap()` → convert to a custom `FruitException`

9. Reactive Number Processor

Create a reactive pipeline that:

1. Starts with `Flux.range(1, 20)`
2. Filters only even numbers
3. Uses `map()` to multiply by 10
4. Uses `flatMap()` to create a small delay (simulating async work)
5. Skips first 2 items
6. Collects final result as List using `collectList()`
7. Handles any error with `onErrorReturn(List.of(999))`

8. Uses proper lifecycle hooks to show start, each item, and completion

Print the final list.